q-zeta reciprocity for preorder lattice-point posets

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Abstract

We prove Conjecture 4.9 of Athanasiadis and Chapoton for every finite preorder. A support refinement of Postnikov's lattice-point formula identifies certain strict filter counts with maximal lattice points. Weighted Ehrhart reciprocity then gives the proposed q-zeta value.

Task 1. Full solution

Let τ be a preorder on E, with $|E| = n \geq 1$. For a subset A write $x(A) = \sum_{e \in A} x_e$, and let $U_\tau(A)$ and $D_\tau(A)$ be its upward and downward closures. Define

$$\mathcal{Q}_\tau = \{x \in \mathbb{R}^E_{\geq 0} : x(I) \leq |I| \text{ for every order ideal } I\}, \qquad P_\tau = \mathcal{Q}_\tau \cap \mathbb{Z}^E,$$

with componentwise order and rank $|a| = a(E)$. This is a downward closed set, so $\operatorname{cov}(a) = |\operatorname{supp}(a)|$. Here cov counts elements covered by a, and $\operatorname{supp}(a) = \{e : a_e > 0\}$. For an indeterminate q, the q-zeta polynomial is characterized by

$$\mathcal{Z}_q(P, [m]_q) = \sum_{a_1 \leq \cdots \leq a_{m-1}} q^{|a_1| + \cdots + |a_{m-1}|}, \qquad [m]_q = \frac{q^m - 1}{q - 1}, \quad m \geq 2.$$

Theorem 1. *For every finite preorder τ,*

$$\boxed{\mathcal{Z}_q(P_\tau, [-1]_q) = (-1)^n \sum_{a \in \operatorname{Max}(P_\tau)} q^{-\operatorname{cov}(a)}.}$$

Consequently $\mathcal{Z}(P_\tau, -1) = (-1)^n |\operatorname{Max}(P_\tau)|$.

Weighted polytopes and support counts

For nonnegative integral weights $y = (y_e)$ and a preorder σ, put

$$\mathcal{Q}_\sigma(y) = \sum_{e \in E} y_e \operatorname{conv}(0, \mathbf{e}_f : f \in U_\sigma(e)).$$

Its inequality description is

$$x \geq 0, \qquad x(I) \leq y(I) \quad \text{for every order ideal } I \text{ of } \sigma. \tag{1}$$

Indeed, a nonnegative vector can be supplied by these simplices exactly when $x(A) \leq y(D_\sigma(A))$ for all $A \subseteq E$, by the transportation form of max-flow/min-cut. The ideal inequalities imply this

because $x(A) \le x(D_\sigma(A))$; conversely take $A = I$. For integer demands and supplies, the flow can be chosen integral. In particular every point of P_τ is a sum of n choices, one from each set $\{0\} \cup \{\mathbf{e}_f : f \in U_\tau(e)\}$. If $|a| < n$, an unused choice can be changed from 0 to $\mathbf{e}_e$. It follows that

$$\mathrm{Max}(P_\tau) = \{a \in P_\tau : |a| = n\}. \tag{2}$$

The established Postnikov formula, in the form of [1, Theorem 2.1 and Lemma 4.1], gives the polynomial

$$F_\sigma(y) = \sum_{a \in P_{\sigma^*}} \prod_{e \in E} \binom{y_e + a_e - 1}{a_e} = |\mathcal{Q}_\sigma(y) \cap \mathbb{Z}^E| \tag{3}$$

for nonnegative integral y. Indeed, use the source's bipartite graph on $E \sqcup \{0\}$, coefficient zero for its universal summand, and coefficient y_e for the summand indexed by e. Its draconian vectors satisfy $a_0 + |a| = n$ and $a \in P_{\sigma^*}$; the factor from the universal summand is $\binom{a_0}{a_0} = 1$. Binomial coefficients here and below have fixed nonnegative lower argument and are interpreted polynomially in the upper argument.

For $S \subseteq E$, define the finite number

$$N_\tau(S) = \#\{b \in \mathbb{Z}^E_{\ge 0} : b(J) \le |J \cap S| - 1 \text{ for every nonempty filter } J \text{ of } \tau\}.$$

Lemma 1. *For every $S \subseteq E$,*

$$N_\tau(S) = \#\{a \in \mathrm{Max}(P_\tau) : \mathrm{supp}(a) \subseteq S\}.$$

Proof. The face of $\mathcal{Q}_\tau$ on which $|x| = n$ and $\mathrm{supp}(x) \subseteq S$ is the Minkowski sum

$$B_\tau(S) = \sum_{e \in E} \mathrm{conv}(\mathbf{e}_s : s \in S \cap U_\tau(e)).$$

This follows directly from its simplex representation: attaining total sum n forces every summand onto its nonzero face, and nonnegative coordinates outside S must all vanish. If S is empty or one of these summands has an empty vertex set, there are no such maximal points. Also $N_\tau(S) = 0$: respectively use the filter E, or a principal filter $U_\tau(e)$ disjoint from S.

Otherwise apply [1, Theorem 2.1] on ground set S with summands $I_e = S \cap U_\tau(e)$ of coefficient one, and an additional universal summand $I_0 = S$ of coefficient zero. No vertex is isolated. Its draconian vectors are the nonnegative vectors (a_0, b) with

$$a_0 + |b| = |S| - 1, \qquad b(A) \le |S \cap U_\tau(A)| - 1 \quad (\varnothing \ne A \subseteq E). \tag{4}$$

Constraints involving the universal summand follow from the total sum. All factors in the counting formula equal one, so lattice points of $B_\tau(S)$ are counted by these vectors. Taking $A = E$ shows that a_0 is determined and nonnegative. The other inequalities are equivalent to those defining $N_\tau(S)$: take $A = J$ for necessity; for sufficiency apply the filter inequality to $U_\tau(A)$ and use $b(A) \le b(U_\tau(A))$. Now use (2). □

Apply ordinary Ehrhart reciprocity to $\mathcal{Q}_{\tau^*}(y)$ with $y = \mathbf{1} + \mathbf{1}_S$. This is a full dimensional lattice polytope, since it contains $[0,1]^E$. Its Ehrhart polynomial in k is $F_{\tau^*}(ky)$, by (3). At $k = -1$ reciprocity gives $F_{\tau^*}(-y) = (-1)^n \#(\mathrm{int}\, \mathcal{Q}_{\tau^*}(y) \cap \mathbb{Z}^E)$. The nonzero normals in (1) show that its interior integer points satisfy $x_e \ge 1$ and $x(J) \le y(J) - 1$ for every nonempty filter J of τ. Subtracting $\mathbf{1}$ therefore proves

$$F_{\tau^*}(-\mathbf{1} - \mathbf{1}_S) = (-1)^n N_\tau(S). \tag{5}$$

Evaluation of the q-zeta polynomial

Fix the last element a of a multichain. Since $[0, a]$ is a product of ordinary chains, the Gaussian binomial formula gives

$$\mathcal{Z}_q(P_\tau, [m]_q) = \sum_{a \in P_\tau} \prod_{e \in E} q^{a_e} \begin{bmatrix} m + a_e - 2 \\ a_e \end{bmatrix}_q . \tag{6}$$

For each integer $a \geq 0$, the factor on the right is

$$q^a \prod_{j=1}^{a} \frac{q^{m+j-2} - 1}{q^j - 1}.$$

It is polynomial in q^m, hence in $[m]_q$, over $\mathbb{Q}(q)$. Thus (6), initially a counting identity for $m \geq 2$, is an identity of polynomials in $[m]_q$. In particular we may substitute $q^m = q^{-1}$, corresponding to $[-1]_q = -q^{-1}$. Set $p = q^{-1}$. The resulting factor is

$$f_a(p) = \begin{cases} 1 & a = 0, \\ -(1+p) & a = 1, \\ p & a = 2, \\ 0 & a \geq 3, \end{cases} \qquad f_a(p) = (1-p)\binom{a-2}{a} + p\binom{a-3}{a}.$$

Expanding the product of the last two-term expression and using (3) yields

$$\begin{aligned} \mathcal{Z}_q(P_\tau, [-1]_q) &= \sum_{S \subseteq E} p^{|S|}(1-p)^{n-|S|} F_{\tau^*}(-\mathbf{1} - \mathbf{1}_S) \\ &= (-1)^n \sum_{S \subseteq E} p^{|S|}(1-p)^{n-|S|} \#\{a \in \operatorname{Max}(P_\tau) : \operatorname{supp}(a) \subseteq S\}, \end{aligned}$$

by (5) and Lemma 1. For a fixed maximal point with support T, the sum of its weights is

$$\sum_{S \supseteq T} p^{|S|}(1-p)^{n-|S|} = p^{|T|}\big(p + (1-p)\big)^{n-|T|} = p^{|T|}.$$

This proves the claimed q-identity because $|T| = \operatorname{cov}(a)$. At $q = 1$ the Gaussian factors in (6) specialize to the ordinary binomial chain-counting polynomials, giving the ordinary zeta assertion. For $n = 0$, the singleton poset has both sides equal to one, so the empty case also holds.

Verification and attribution

The conjecture and the weighted counting input are due to Athanasiadis–Chapoton and Postnikov. The support lemma and its combination with weighted reciprocity supply the proof. Source Remark 4.10 already derives the ordinary specialization conditionally from the descent conjecture; the proof here establishes the full q-refinement directly.

All 389 labeled preorders of sizes 1–4 pass exact polynomial checks, and all 5,930 support subsets pass the support and reciprocity counts. An independent weighted-chain dynamic program supplies 2,035 values and 423 exact interpolations. Larger examples and the source's five-element example also pass. All 26 pages of the current primary were read; bounded checks identified no prior full target claim or target journal final. Review is by the authoring agent only.

Tasks 2–5. Proposed follow-up directions

Task 2. Construct an explicit support-preserving bijection underlying Lemma 1, replacing equality of Postnikov counts.

Task 3. Determine which integral polymatroids satisfy the same q-zeta reciprocity identity with their maximal elements.

Task 4. Retain separate variables for each coordinate in the support lemma and seek a multivariate zeta specialization realizing them.

Task 5. Find combinatorial interpretations for $\mathcal{Z}_q(P_\tau, [-r]_q)$ at other positive integers r, extending the case $r = 1$.

References

[1] C. A. Athanasiadis and F. Chapoton, *Polytopes and posets associated to preorders*, arXiv:2605.26916v1, 2026.